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AIM of the Experiment :-

The design, implement and test the given Boolean function $Y = AB + C$ using
(i) NAND Gates (ii) NOR Gates

Objective :-

To verify implementation of the given boolean function $Y = AB + C$ using NAND Gates only and NOR Gates only.

Apparatus Required :-

Sl. No	Name of the Components	Specification	Quantity
1	Trainer Kit	XPO-ANALOG II	1
2.	Bread Board	-	1
3.	NAND Gate (IC 74LS00)	Quad 2 input NAND Gate	1
4.	NOR Gate (IC 74LS02)	Quad 2 input NOR Gate	1
5.	Connecting wires	—	As per Required

Theory :-

NAND and NOR Gates are known as universal gates because any boolean function can be implemented using only NAND and NOR gates only

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$$AB+C = \overline{\overline{AB+C}} = \overline{\overline{AB}\cdot\overline{C}}$$
$$= \overline{\overline{AB}}+\overline{\overline{C}}$$
$$= AB+C$$

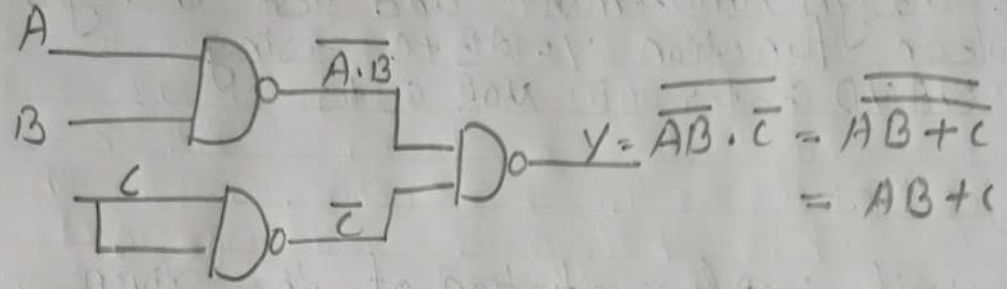


Fig 2.1: Implementation of $Y = AB+C$ using NAND Gates only

$$\overline{\overline{AB}+C} = \overline{\overline{AB}}+\overline{\overline{C}}$$
$$= \overline{\overline{A}+\overline{B}}+\overline{\overline{C}}$$
$$= \overline{A}+\overline{B}+C$$

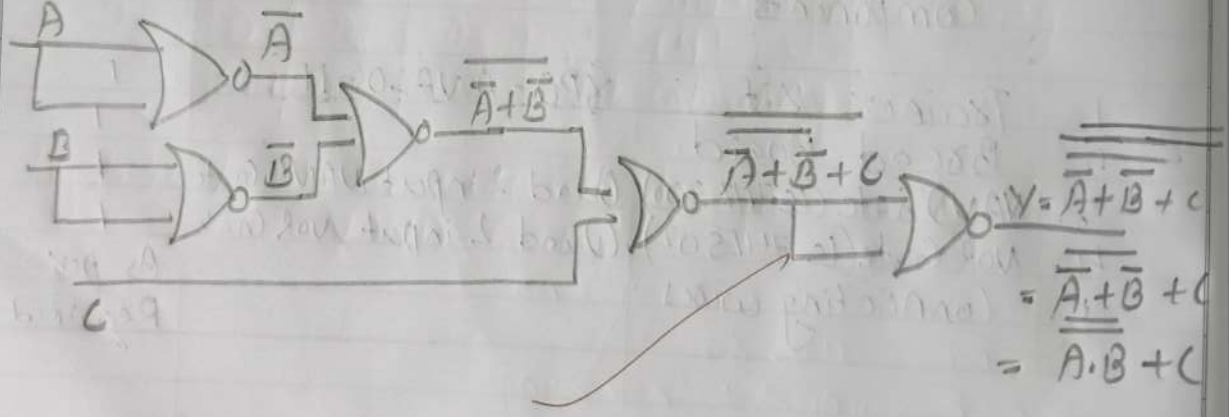


Fig 2.2: Implementation of $Y = AB+C$ using NOR Gates only

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The Boolean function is $Y = AB + C$ using Boolean algebra and De. morgan's law the given Boolean function can be implemented using NAND and NOR gates only.

Truth Table for the given Boolean function

Table 2.1

A	B	C	AB	Y = AB + C
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	1

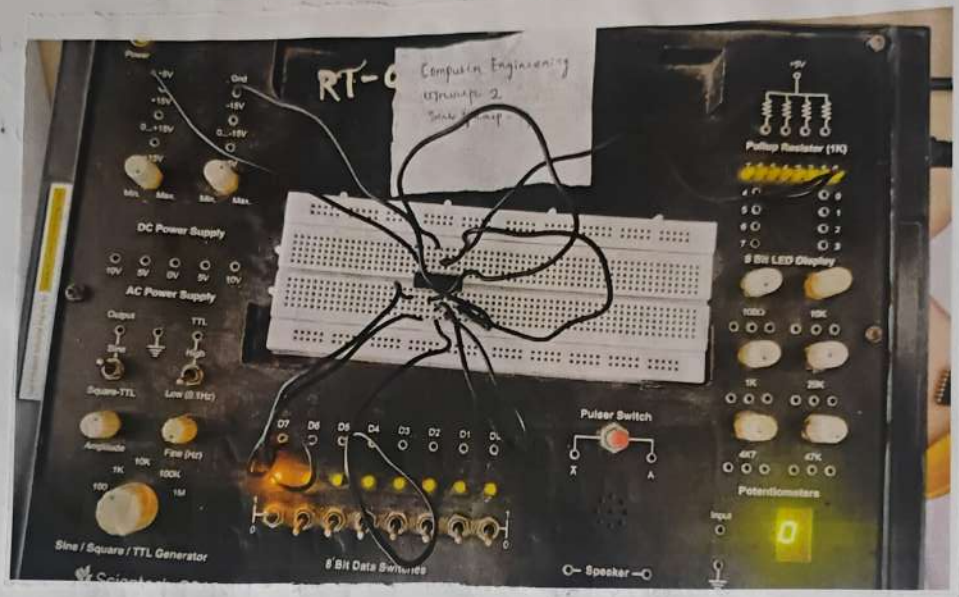
Procedure:-

A. Using NAND Gates only:-

1. Implementation Boolean function ($Y = AB + C$) using NAND gates according to the derived logic circuit.

2. Connect required NAND gate IC to the breadboard

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Input			Output
A	B	C	Y=AB+C
D_7	D_6	D_5	(4)
ON	OFF	OFF	Not Glowing

Fig 2.3: Y=AB+C Circuit design using NAND Gates.

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3. Apply all possible combination of inputs (A) (B), and (C).

4. Observe and record the output (Y).

B. Using NOR Gates only:-

1. Implement the Boolean function (Y= AB+C) using NOR gates according to the derived logic circuit.

2. Connect the required NOR gate to the breadboard.

3. Apply all possible combination of input (A) (B) & (C)

4. Observe and record the output (Y).

Observation

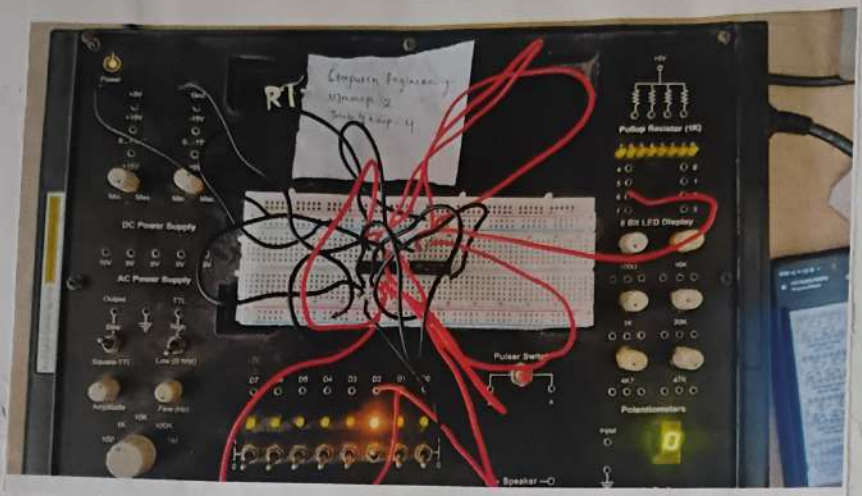
① Using NAND Gates only

Observation Table 1

Input			Output	
A	B	C	Y= AB+C	Remark
(D_7)	(D_6)	(D_5)	(L_4)	
OFF	OFF	OF	Not Glowing	pratical output matching with theortical output
OFF	OFF	ON	Glowing	
OFF	ON	OF	Not Glowing	

ON → 1, OFF → 0

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Input			Output
A (D_2)	B (D_1)	C (D_0)	Y = AB+C (6)
ONN	OFF	OFF	Not Glowing

Fig 2.4: Y = AB+C Circuit design using NOR gates.

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Observation Table 1

Input			Output	
A	B	C	Y=AB+C	Remark
(D_7)	(D_6)	(D_5)	(LI)	
OFF	ON	ON	Glowing	pratical output matching with theoritical output
ON	OFF	OFF	Not Glowing	
ON	OFF	ON	Glowing	
ON	ON	OFF	Glowing	
ON	ON	ON	Glowing	

ON → 1, OFF → 0

(ii) Using NOR Gates only

Observation Table 2

Input			Output	
A(D_3)	B(D_2)	C(D_1)	Y=AB+C(D_0)	Remark
OFF	OFF	OFF	Not Glowing	
OFF	OFF	ON	Glowing	
OFF	ON	OFF	Not Glowing	
OFF	ON	ON	Glowing	Pratical output matching with the theoritical output
ON	OFF	OFF	Not Glowing	
ON	OFF	ON	Glowing	
ON	ON	OFF	Glowing	
ON	ON	ON	Glowing	

ON → 1, OFF → 0

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Conclusion:-

The Boolean function $Y = AB + C$ has been successfully implemented using NAND Gates only and NOR gates only. The pratical outputs have been varified for all the possible combination of inputs and found to be in agreement with the theoritical truth table.

Name- Sambit Kumar Jena
Regd no- 2511 0283
Branch- Computer Engineering
Group- A2
Sub Group- 2

Teacher's Signature JNesra 08/26